



MUHAMMAD TAHA

22F-3277

COAL LAB



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MASM Task with Debugger

Task 1: what is Origin? [3]

Origin is the memory address at which a program or a specific section of code should be loaded or executed. It is the starting point of the program.

Task 2: Calculate Physical Address with the Following values. [3]

1) DS = 073F and SI = 0030

Ans = 07420

2) CS = 08D3 and IP = 0160

Ans = 08E90

Task 3: Display the contents of the defined memory locations 120, 133, 122 using D command and fill following table. [3]

Memory Location	Value
[120]	00
[133]	00
[122]	00

Task 4: Edit the contents of the defined memory locations 120, 133, 122 using e command and also display, also fill following table. [3]

Memory Location	Value
[120]	27
[133]	28
[122]	29

Task 5: Subtract the content of 120 location by 133 and then store the result in the 120 location. [3]

```
mov ax, [120]
mov bx, [133]
sub bx, ax
mov [120], ax
```

NASM Tasks

Task1: Write assembly language code to add 2 number and also fill the following table: [5]

Assembly Code	Machine Code	No. of Bytes	AX	BX	CS	IP
Mov AX,0005	B80500	3	0005	0000	19F5	0103
Mov BX,000A	BB0A00	3	0005	000A	19F5	0106
Mov AX,BX	89D8	2	000A	000A	19F5	0108
Mov AX,4C00	B8004C	3	4C00	000A	19F5	010B

Task2: Write assembly language code to subtract 2 number and also fill the following table: [5]

Assembly Code	Machine Code	No. of Bytes	AX	BX	CS	IP
Mov AX,0005	B80500	3	0005	0000	19F5	0103
MOV BX,000A	BB0A00	3	0005	000A	19F5	0106
SUB AX,BX	29D8	2	FFFB	000A	19F5	0108
AX,4C00	B8004C	3	4C00	000A	19F5	010B

Task3: Write assembly language code to swap 2 number and also fill the following table: [5]

Assembly Code	Machine Code	No. of Bytes	AX	BX	CX	CS	IP
mov ax, 0005	B80500	3	0005	0000	0011	19F5	0103
mov bx, 000A	BB0A00	3	0005	000A	0011	19F5	0106
mov cx,ax	89C1	2	0005	000A	0005	19F5	0108
mov ax,bx	89D8	2	000A	000A	0005	19F5	010A
mov bx,cx	89CB	2	000A	0005	0005	19F5	010C